LAB 2 and 3

Objective:

1. **Write an assembly language program for the addition of two 16-bit hex numbers.**
2. **Write an assembly language program for the subtraction of two 16-bit hex numbers.**
3. **Write an assembly language program for the multiplication of two 8-bit hex numbers.**
4. **Write an assembly language program for the division of two 8-bit hex numbers.**

Program to add two 16-bit numbers.

Statement: Add numbers 1124H & 2253H and store result in memory location 2055H & 2056H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	21	LXI H, 1020	Load 16-bit data 1020H to HL pair
C001	20		
C002	10		
C003	11	LXI D, 1121	Load 16-bit data 1121H to HL pair
C004	21		
C005	11		
C006	7D	MOV A, L	Move contents of register L to Accumulator
C007	83	ADD E	Add contents of Accumulator and E register
C008	6F	MOV L, A	Move contents of Accumulator to L register
C009	7C	MOV A, H	Move contents of register H to Accumulator
C00A	8A	ADC D	Add contents of Accumulator and D register with carry
C00B	67	MOV H, A	Move contents of Accumulator to register H
C00C	22	SHLD C050	Store contents of HL pair in memory address C050H & C051H
C00D	50		
C00E	C0		
C00F	76	HLT	Terminate the program.

Input:

Register Pair Data

HL 1020H

DE 1121H

Output:

Memory Location Data

C050H 41H

C051H 21H

Program to add two 16-bit numbers.

Statement: Input first number from memory location C050H & C051H and second number from memory location C052H & C053H and store result in memory location C055H & C056H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	2A	LHLD C052	Load 16-bit number from memory location C052H & C053H to HL pair

C001	52		
C002	C0		
C003	EB	XCHG	Exchange contents of HL pair and DE pair
C004	2A	LHLD C050	Load 16-bit number from memory location C050H & C051H to HL
C005	50		
C006	C0		
C007	7D	MOV A, L	Move contents of register L to Accumulator
C008	83	ADD E	Add contents of Accumulator and E register
C009	6F	MOV L, A	Move contents of Accumulator to L register
C00A	7C	MOV A, H	Move contents of register H to Accumulator
C00B	8A	ADC D	Add contents of Accumulator and D register with carry
C00C	67	MOV H,A	Move contents of Accumulator to register H
C00D	22	SHLD C055	Store contents of HL pair in memory address 2055H & 2056H
C00E	55		
C00F	C0		
C010	76	HLT	Terminate the program.

Input:

Memory Location Data

C050H 33H

C051H 45H

C052H 24H

C053H 34H

Output:

Memory Location Data

C055H 57H

C056H 79H

Program to subtract two 16-bit numbers.

Statement: Subtract number 1234H from 4897H and store result in memory location C055H & C056H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	21	LXI H, 4897	Load 16-bit data 4897H to HL pair
C001	97		
C002	48		
C003	11	LXI D, 1234	Load 16-bit data 1234H to DE pair
C004	34		
C005	12		
C006	7D	MOV A, L	Move contents of register L to Accumulator
C007	93	SUB E	Subtract contents of Accumulator and E register
C008	6F	MOV L, A	Move contents of Accumulator to L register
C009	7C	MOV A, H	Move contents of register H to Accumulator
C00A	9A	SBB D	Subtract contents of Accumulator and D register with borrow

C00B	67	MOV H, A	Move contents of Accumulator to register H
C00C	22	SHLD C055	Store contents of HL pair in memory address 2055H & 2056H
C00D	50		
C00E	C0		
C00F	76	HLT	Terminate the program.

Input:

Register Pair Data

HL 4897H

DE 1234H

Output:

Memory Location Data

C055H 63H

C056H 36H

Program to subtract two 16-bit numbers.

Statement: Input first number from memory location C050H & C051H and second number from memory location C052H & C053H and store result in memory location C055H & C056H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	2A	LHLD C052	Load 16-bit number from memory location C052H & C053H to HL pair
C001	52		
C002	C0		
C003	EB	XCHG	Exchange contents of HL pair and DE pair
C004	2A	LHLD C050	Load 16-bit number from memory location C050H & C051H to HL
C005	50		
C006	C0		
C007	7D	MOV A, L	Move contents of register L to Accumulator
C008	93	SUB E	Subtract contents of Accumulator and E register
C009	6F	MOV L, A	Move contents of Accumulator to L register
C00A	7C	MOV A, H	Move contents of register H to Accumulator
C00B	9A	SBB D	Subtract contents of Accumulator and D register with Borrow
C00C	67	MOV H,A	Move contents of Accumulator to register H
C00D	22	SHLD C055	Store contents of HL pair in memory address C055H & C056H
C00E	55		
C00F	C0		
C010	76	HLT	Terminate the program.

Input:

Memory Location Data

C050H 78H

C051H 45H

C052H 24H

C053H 34H

Output:

Memory Location Data

C055H 54H

C056H 11H

Program to multiply two 8-bit numbers.

Statement: Multiply 06 and 03 and store result in memory location C055H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3E	MVI A, 00	Loads the accumulator with 00H hex number
C001	00		
C002	06	MVI B, 06	Loads the accumulator with 06H hex number
C003	06		
C004	0E	MVI C, 03	Loads the accumulator with 03H hex number
C005	03		
C006	80	X:ADD B	Add contents of Accumulator and B register
C007	0D	DCR C	The contents of register C is decremented by 1
C008	C2	JNZ X	Jump if no zero to X
C009	06		
C00A	00		
C00B	32	STA C055	Store result in memory location C055H
C00C	55		
C00D	C0		
C00E	76	HLT	Terminate the program

Input: B = 06H C = 03H **Output:** C055 = 12H

Write an Assembly language program multiply two 8 bit numbers stored at address C050 and C051. Result is stored at address C055 and C056.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3A	LDA C050	Load contents from memory location C050H to accumulator
C001	50		
C002	C0		
C003	5F	MOV E, A	Move contents of Accumulator to E register
C004	3A	LDA C051	Load contents from memory location C051H to accumulator
C005	51		
C006	C0		
C007	4F	MOV C, A	Move contents of Accumulator to C register
C008	16	MVI D, 00	Loads register D with 00H hex number
C009	00		
C00A	21	LXI H, 0000	Load 16-bit data 0000H to H-L pair
C00B	00		
C00C	00		
C00D	19	X: DAD D	The 16-bit contents of the register pair DE are added to the contents of H-L pair. If the result is larger than 16 bits, then CY is Set.
C00E	0D	DCR C	The contents of register C is decremented by 1
C00F	C2	JNZ X	Jump if no zero to X
C010	0D		

C011	00		
C012	22	SHLD C055	Store contents of HL pair in memory address C055H & C056H
C013	50		
C014	C0		
C015	76	HLT	Terminate the program.

Input:

Memory Location Data

C050H 09H

C051H 0AH

Output:

Memory Location Data

C055H 0AH

C056H 5AH

Program to divide to 8-bit numbers.

Statement: Divide 08H and 03H and store quotient in memory location C055H and remainder in memory location C056H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3E	MVI A, 08	Loads the accumulator with 08H hex number
C001	08		
C002	06	MVI B, 03	Loads the accumulator with 03H hex number
C003	03		
C004	0E	MVI C, 00	Loads the accumulator with 00H hex number
C005	00		
C006	B8	X: CMP B	The contents of the Register B is compared with the contents of the accumulator and both contents are preserved
C007	DA	JC Y	Jump if carry to Y
C008	0F		
C009	00		
C00A	90	SUB B	Subtract contents of Accumulator and B register
C00B	0C	INR C	The contents of register C is incremented by 1
C00C	C3	JMP X	Jump unconditionally
C00D	06		
C00E	00		
C00F	32	Y: STA C056	Store remainder in memory location C056H
C010	56		
C011	C0		
C012	79	MOV A, C	Move contents of register C to Accumulator
C013	32	STA C055	Store quotient in memory location C056H
C014	55		
C015	C0		
C016	76	HLT	Terminate the program.

Output: Quotient: C056 = 02H Remainder:C055 = 02H